

# Peng Zhao

pz2110@nyu.edu  
+86 189-2626-9300

## Education

### New York University – Center for Data Science

New York, NY, USA

M.S. in Data Science, GPA: 3.94 / 4

2025.09 – 2027.05

### University of California, Santa Barbara

Santa Barbara, CA, USA

B.S. in Statistics and Data Science, GPA: 3.92 / 4, High Honors

2021.09 – 2025.08

Exchange Study, University of Manchester, Spring 2025

## Research Interests

Multimodal Learning; Efficient Transformers; Information-theoretic Learning; Scientific Foundation Models

## Research Experience

### Urban Systems Lab, NYU

New York, NY, USA

**Graduate Research Assistant**, Supervisor: Madhavi Jain

2025.11 – Present

ClimatelQ Project: Generative Surrogate Modeling for Urban Climate Downscaling [[website](#)]

- Lead ML development for ClimatelQ's atmospheric component, building neural surrogates for WRF physical simulator downscaling from 100-km CMIP6 data to 500-m urban temperature fields.
- Reframed direct downscaling as a physically grounded 100 km → 12.5 km → 2.5 km → 500 m cascade super-resolution, yielding better-conditioned intermediate learning problems.
- Developed flow-matching models with U-Net architectures and efficient conditioning paths that inject spatiotemporal climate fields, static urban features, and temporal metadata.
- Ablations showed that nested models trained more reliably with smaller capacity and produced sharper fields, clear land–water boundaries, and diverse local urban structure without observed mode collapse.

### Climate Hazards Center, UCSB

Santa Barbara, CA, USA

**Research Assistant**, Supervisor: Shraddhanand Shukla

2024.06 – 2025.03

- Investigated ML daily precipitation estimates as an independent reference for validating CHIRPS datasets in data-sparse regions, using well-observed North America as the training domain.
- Identified challenges from zero-inflated, heavy-tailed precipitation data, including ill-defined normalization, regression bias, and failure to recover rare extremes.
- Evaluated cold-cloud temperature and soil-moisture as external signals; concluded that cold-cloud temperature was informative under MSE and 5-mm wet/dry classification metrics.

### Data Science Capstone Researcher [[poster](#)]

2023.09 – 2024.06

- Collaborated in a five-person team to evaluate four CHIRPS 3.0 daily precipitation products against independent station and satellite data, including spatial and lagged soil-moisture analyses.
- Developed a conditional GAN proof of concept for daily precipitation reconstruction from multiday aggregated data; co-presented the study at UCSB's capstone poster session.

## Projects

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### **Prompt-Conditioned Residual Re-aggregation for Lightweight Domain Adaptation** [\[paper\]](#) [\[code\]](#)

Course Project, Machine Learning, NYU

2026.01 – 2026.05

- Initiated a parameter-efficient adaptation study asking whether prompt-conditioned residual control could improve frozen language models across heterogeneous text domains.
- Developed layerwise write-strength scaling and lower-triangular residual re-aggregation variants, using mean-pooled and last-token final-layer latents as a fixed-length prompt representation.
- On frozen Qwen3-0.6B with mixed WikiText–PubMed data, prompt-conditioned re-aggregation reduced test perplexity from 16.81 to 15.23 (9.4%); ablations attributed most gains to re-aggregation.
- Designed controller diagnostics to inspect collapse via cosine similarities; re-aggregation preserved better prompt-dependent variation and showed non-trivial domain separation.

### **Layerwise Signals for Transformer MLP Compression** [\[paper\]](#) [\[slide\]](#)

Course Project, Information Theory, NYU

2026.01 – 2026.05

- Proposed a complexity–importance hypothesis for allocating low-rank compression budgets across transformer MLP layers using activation effective ranks and trace.
- Built an end-to-end pipeline for layer probing, activation-aware factorization, rank allocation, compression, and perplexity evaluation, comparing four language models from 0.6B to 1.7B sizes.
- Found a surprisingly consistent inverse relation between centered effective rank and MLP-output trace: early higher-dimensional representations often contributed less strongly in magnitude to the residual stream.
- Activation-aware factorization substantially outperformed truncated SVD, but nonuniform rank allocation did not beat a uniform baseline, showing that layer statistics alone did not predict compression sensitivity.

## Selected Coursework

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**Mathematics & Probability:** Methods of Analysis; Numerical Analysis; Stochastic Processes; Optimization and Computational Linear Algebra; Inference and Representation (ongoing)

**Statistics & Machine Learning:** Bayesian Data Analysis; Time Series; Statistical Machine Learning; Information Theory and Cognition; Fundamentals of NLP (ongoing)

## Technical Skills

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**Programming:** Python; R; SQL

**Data Science:** PyTorch; Hugging Face Transformers; Apache Spark; Dask

**Tools:** Slurm; GPU cluster workflows; Git

**GRE:** 329 (V159, Q170)